Neural Computation -- 1996

96 article(s), all issues

1. Binocular receptive field models, disparity tuning, and characteristic disparity

ZhuYu-Dong, QianNing

https://doi.org/10.5555/1362517.1362520

Disparity tuning of visual cells in the brain depends on the structure of their binocular receptive fields (RFs). Freeman and coworkers have found that binocular RFs of a typical simple cell can be...

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2. Hebbian learning of context in recurrent neural networks

BrunelNicolas

https://doi.org/10.5555/1362517.1362523

Single electrode recordings in the inferotemporal cortex of monkeys during delayed visual memory tasks provide evidence for attractor dynamics in the observed region. The persistent elevated delay ...

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3. Lower Bounds for the Computational Power of Networks of Spiking Neurons

Wolfgang Maass

https://doi.org/10.1162/neco.1996.8.1.1 | pp. 1-40

We investigate the computational power of a formal model for networks of spiking neurons. It is shown that simple operations on phase differences between spike-trains provide a very powerful computational tool that can in principle be used to carry out highly complex computations on a small network of spiking neurons. We construct networks of spiking neurons that simulate arbitrary threshold circuits, Turing machines, and a certain type of random access machines with real valued inputs. We also show that relatively weak basic assumptions about the response and threshold functions of the spiking neurons are sufficient to employ them for such computations.

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4. A Short Proof of the Posterior Probability Property of Classifier Neural Networks

Raúl Rojas

https://doi.org/10.1162/neco.1996.8.1.41 | pp. 41-43

It is now well known that neural classifiers can learn to compute a posteriori probabilities of classes in input space. This note offers a shorter proof than the traditional ones. Only one class has to be considered and straightforward minimization of the error function provides the main result. The method can be extended to any kind of differentiable error function. We also present a simple visual proof of the same theorem, which stresses the fact that the network must be perfectly trained and have enough plasticity.

5. Coding of Time-Varying Signals in Spike Trains of Integrate-and-Fire Neurons with Random Threshold

Fabrizio Gabbiani, Christof Koch

https://doi.org/10.1162/neco.1996.8.1.44 | pp. 44-66

Recently, methods of statistical estimation theory have been applied by Bialek and collaborators (1991) to reconstruct time-varying velocity signals and to investigate the processing of visual information by a directionally selective motion detector in the fly's visual system, the H1 cell. We summarize here our theoretical results obtained by studying these reconstructions starting from a simple model of H1 based on experimental data. Under additional technical assumptions, we derive a closed expression for the Fourier transform of the optimal reconstruction filter in terms of the statistics of the stimulus and the characteristics of the model neuron, such as its firing rate. It is shown that linear reconstruction filters will change in a nontrivial way if the statistics of the signal or the mean firing rate of the cell changes. Analytical expressions are then derived for the mean square error in the reconstructions and the lower bound on the rate of information transmission that was estimated experimentally by Bialek et al. (1991). For plausible values of the parameters, the model is in qualitative agreement with experimental data. We show that the rate of information transmission and mean square error represent different measures of the reconstructions: in particular, satisfactory reconstructions in terms of the mean square error can be achieved only using stimuli that are matched to the properties of the recorded cell. Finally, it is shown that at least for the class of models presented here, reconstruction methods can be understood as a generalization of the more familiar reverse-correlation technique.

6. A Simple Spike Train Decoder Inspired by the Sampling Theorem

David A. August, William B. Levy

https://doi.org/10.1162/neco.1996.8.1.67 | pp. 67-84

Reconstructing a time-varying stimulus estimate from a spike train (Bialek's "decoding" of a spike train) has become an important way to study neural information processing. In this paper, we describe a simple method for reconstructing a time-varying current injection signal from the simulated spike train it produces. This technique extracts most of the information from the spike train, provided that the input signal is appropriately matched to the spike generator. To conceptualize this matching, we consider spikes as instantaneous "samples" of the somatic current. The Sampling Theorem is then applicable, and it suggests that the bandwidth of the injected signal not exceed half the spike generator's average firing rate. The average firing rate, in turn, depends on the amplitude range and DC bias of the injected signal. We hypothesize that nature faces similar problems and constraints when transmitting a time-varying waveform from the soma of one neuron to the dendrite of the postsynaptic cell.

7. A Model of Spatial Map Formation in the Hippocampus of the Rat

K. I. Blum, L. F. Abbott

https://doi.org/10.1162/neco.1996.8.1.85 | pp. 85-93

Using experimental facts about long-term potentiation (LTP) and hippocampal place cells, we model how a spatial map of the environment can be created in the rat hippocampus. Sequential firing of place cells during exploration induces, in the model, a pattern of LTP between place cells that shifts the location coded by their ensemble activity away from the actual location of the animal. These shifts provide a navigational map that, in a simulation of the Morris maze, can guide the animal toward its goal. The model demonstrates how behaviorally generated modifications of synaptic strengths can be read out to affect subsequent behavior. Our results also suggest a way that navigational maps can be constructed from experimental recordings of hippocampal place cells.

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8. A Neural Model of Olfactory Sensory Memory in the Honeybee's Antennal Lobe

Christiane Linster, C. Masson

https://doi.org/10.1162/neco.1996.8.1.94 | pp. 94-114

We present a neural model for olfactory sensory memory in the honeybee's antennal lobe. To investigate the neural mechanisms underlying odor discrimination and memorization, we exploit a variety of morphological, physiological, and behavioral data. The model allows us to study the computational capacities of the known neural circuitry, and to interpret under a new light experimental data on the cellular as well as on the neuronal assembly level. We propose a scheme for memorization of the neural activity pattern after stimulus offset by changing the local balance between excitation and inhibition. This modulation is achieved by changing the intrinsic parameters of local inhibitory neurons or synapses.

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9. A Spherical Basis Function Neural Network for Modeling Auditory Space

Rick L. Jenison, Kate Fissell

https://doi.org/10.1162/neco.1996.8.1.115 | pp. 115-128

This paper describes a neural network for approximation problems on the sphere. The von Mises basis function is introduced, whose activation depends on polar rather than Cartesian input coordinates. The architecture of the von Mises Basis Function (VMBF) neural network is presented along with the corresponding gradient-descent learning rules. The VMBF neural network is used to solve a particular spherical problem of approximating acoustic parameters used to model perceptual auditory space. This model ultimately serves as a signal processing engine to synthesize a virtual auditory environment under headphone listening conditions. Advantages of the VMBF over standard planar Radial Basis Functions (RBFs) are discussed.

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10. On Convergence Properties of the EM Algorithm for Gaussian Mixtures

Lei Xu, Michael I. Jordan

https://doi.org/10.1162/neco.1996.8.1.129 | pp. 129-151

We build up the mathematical connection between the “Expectation-Maximization” (EM) algorithm and gradient-based approaches for maximum likelihood learning of finite gaussian mixtures. We show that the EM step in parameter space is obtained from the gradient via a projection matrix P, and we provide an

explicit expression for the matrix. We then analyze the convergence of EM in terms of special properties of P and provide new results analyzing the effect that P has on the likelihood surface. Based on these mathematical results, we present a comparative discussion of the advantages and disadvantages of EM and other algorithms for the learning of gaussian mixture models.

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11. A Comparison of Some Error Estimates for Neural Network Models

Robert Tibshirani

https://doi.org/10.1162/neco.1996.8.1.152 | pp. 152-163

We discuss a number of methods for estimating the standard error of predicted values from a multilayer perceptron. These methods include the delta method based on the Hessian, bootstrap estimators, and the “sandwich” estimator. The methods are described and compared in a number of examples. We find that the bootstrap methods perform best, partly because they capture variability due to the choice of starting weights.

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12. Neural Networks for Optimal Approximation of Smooth and Analytic Functions

H. N. Mhaskar

https://doi.org/10.1162/neco.1996.8.1.164 | pp. 164-177

We prove that neural networks with a single hidden layer are capable of providing an optimal order of approximation for functions assumed to possess a given number of derivatives, if the activation function evaluated by each principal element satisfies certain technical conditions. Under these conditions, it is also possible to construct networks that provide a geometric order of approximation for analytic target functions. The permissible activation functions include the squashing function $(1 - e^{-x})^{-1}$ as well as a variety of radial basis functions. Our proofs are constructive. The weights and thresholds of our networks are chosen independently of the target function; we give explicit formulas for the coefficients as simple, continuous, linear functionals of the target function.

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13. Equivalence of Linear Boltzmann Chains and Hidden Markov Models

David Mackay

https://doi.org/10.1162/neco.1996.8.1.178 | pp. 178-181

Several authors have studied the relationship between hidden Markov models and “Boltzmann chains” with a linear or “time-sliced” architecture. Boltzmann chains model sequences of states by defining state-state transition energies instead of probabilities. In this note I demonstrate that under the simple condition that the state sequence has a mandatory end state, the probability distribution assigned by a strictly linear Boltzmann chain is identical to that assigned by a hidden Markov model.

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14. Diagrammatic Derivation of Gradient Algorithms for Neural Networks

Eric A. Wan, Françoise Beaufays

https://doi.org/10.1162/neco.1996.8.1.182 | pp. 182-201

Deriving gradient algorithms for time-dependent neural network structures typically requires numerous chain rule expansions, diligent bookkeeping, and careful manipulation of terms. In this paper, we show how to derive such algorithms via a set of simple block diagram manipulation rules. The approach provides a common framework to derive popular algorithms including backpropagation and backpropagation-through-time without a single chain rule expansion. Additional examples are provided for a variety of complicated architectures to illustrate both the generality and the simplicity of the approach.

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15. Does Extra Knowledge Necessarily Improve Generalization?

David Barber, David Saad

https://doi.org/10.1162/neco.1996.8.1.202 | pp. 202-214

The generalization error is a widely used performance measure employed in the analysis of adaptive learning systems. This measure is generally critically dependent on the knowledge that the system is given about the problem it is trying to learn. In this paper we examine to what extent it is necessarily the case that an increase in the knowledge that the system has about the problem will reduce the generalization error. Using the standard definition of the generalization error, we present simple cases for which the intuitive idea of “reducivity”—that more knowledge will improve generalization—does not hold. Under a simple approximation, however, we find conditions to satisfy “reducivity.” Finally, we calculate the effect of a specific constraint on the generalization error of the linear perceptron, in which the signs of the weight components are fixed. This particular restriction results in a significant improvement in generalization performance.

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16. Encoding with Bursting, Subthreshold Oscillations, and Noise in Mammalian Cold Receptors

André Longtin, Karin Hinzer

https://doi.org/10.1162/neco.1996.8.2.215 | pp. 215-255

Mammalian cold thermoreceptors encode steady-state temperatures into characteristic temporal patterns of action potentials. We propose a mechanism for the encoding process. It is based on Plant's ionic model of slow wave bursting, to which stochastic forcing is added. The model reproduces firing patterns from cat lingual cold receptors as the parameters most likely to underlie the thermosensitivity of these receptors varied over a 25 degrees C range. The sequence of firing patterns goes from regular bursting, to simple periodic, to stochastically phase-locked firing or "skipping." The skipping at higher temperatures is shown to necessitate an interaction between noise and a subthreshold endogenous oscillation in the receptor. The basic period of all patterns is robust to noise. Further, noise extends the range of encodable stimuli. An increase in firing irregularity with temperature also results from the loss of stability accompanying the approach by the slow dynamics of a reverse Hopf bifurcation. The results are not dependent on the precise details of the Plant model, but are generic features of models where an autonomous slow wave arises through a Hopf bifurcation. The model also addresses the variability of the firing patterns across fibers. An alternate model of slow-wave bursting (Chay and Fan 1993) in which skipping can occur without noise is also analyzed here in the context of cold thermoreception. Our study quantifies the possible origins and relative contribution of deterministic and stochastic dynamics to the coding scheme. Implications of our

findings for sensory coding are discussed.

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17. Associative Memory with Uncorrelated Inputs

Ronald Michaels

https://doi.org/10.1162/neco.1996.8.2.256 | pp. 256-259

In hybrid learning schemes a layer of unsupervised learning is followed by supervised learning. In this situation a connection between two unsupervised learning algorithms, principal component analysis and decorrelation, and a supervised learning algorithm, associative memory, is shown. When associative memory is preceded by principal component analysis or decorrelation it is possible to take advantage of the lack of correlation among inputs to associative memory to show that correlation matrix memory is a least squares solution to the supervised learning problem.

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18. Statistical Independence and Novelty Detection with Information Preserving Nonlinear Maps

Lucas C. Parra, Gustavo Deco, Stefan Miesbach

https://doi.org/10.1162/neco.1996.8.2.260 | pp. 260-269

According to Barlow (1989), feature extraction can be understood as finding a statistically independent representation of the probability distribution underlying the measured signals. The search for a statistically independent representation can be formulated by the criterion of minimal mutual information, which reduces to decorrelation in the case of gaussian distributions. If nongaussian distributions are to be considered, minimal mutual information is the appropriate generalization of decorrelation as used in linear Principal Component Analyses (PCA). We also generalize to nonlinear transformations by only demanding perfect transmission of information. This leads to a general class of nonlinear transformations, namely symplectic maps. Conservation of information allows us to consider only the statistics of single coordinates. The resulting factorial representation of the joint probability distribution gives a density estimation. We apply this concept to the real world problem of electrical motor fault detection treated as a novelty detection task.

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19. Neural Network Models of Perceptual Learning of Angle Discrimination

Germán Mato, Haim Sompolinsky

https://doi.org/10.1162/neco.1996.8.2.270 | pp. 270-299

We study neural network models of discriminating between stimuli with two similar angles, using the two-alternative forced choice (2AFC) paradigm. Two network architectures are investigated: a two-layer perceptron network and a gating network. In the two-layer network all hidden units contribute to the decision at all angles, while in the other architecture the gating units select, for each stimulus, the appropriate hidden units that will dominate the decision. We find that both architectures can perform the task reasonably well for all angles. Perceptual learning has been modeled by training the networks to perform the task, using unsupervised Hebb learning algorithms with pairs of stimuli at fixed angles theta and delta theta. Perceptual transfer is studied by measuring the performance of the network on stimuli with theta' not equal to theta. The two-layer perceptron shows a partial transfer for angles that are within a distance a from theta, where a is

the angular width of the input tuning curves. The change in performance due to learning is positive for angles close to theta, but for magnitude of theta-theta' approximately a it is negative, i.e., its performance after training is worse than before. In contrast, negative transfer can be avoided in the gating network by limiting the effects of learning to hidden units that are optimized for angles that are close to the trained angle.

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20. Directional Filling-in

Karl Frederick Arrington

https://doi.org/10.1162/neco.1996.8.2.300 | pp. 300-318

The filling-in theory of brightness perception has gained much attention recently owing to the success of vision models. However, the theory and its instantiations have suffered from incorrectly dealing with transitive brightness relations. This paper describes an advance in the filling-in theory that overcomes the problem. The advance is incorporated into the BCS/FCS neural network model, which allows it, for the first time, to account for all of Arend's test stimuli for assessing brightness perception models. The theory also suggests a new teleology for parallel ON- and OFF-channels.

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21. Binary-Oscillator Networks: Bridging a Gap between Experimental and Abstract Modeling of Neural Networks

Weiping Wang

https://doi.org/10.1162/neco.1996.8.2.319 | pp. 319-339

This paper proposes a simplified oscillator model, called binary-oscillator, and develops a class of neural network models having binary-oscillators as basic units. The binary-oscillator has a binary dynamic variable v = +/- 1 modeling the "membrane potential" of a neuron, and due to the presence of a "slow current" (as in a classical relaxation-oscillator) it can oscillate between two states. The purpose of the simplification is to enable abstract algorithmic study on the dynamics of oscillator networks. A binary-oscillator network is formally analogous to a system of stochastic binary spins (atomic magnets) in statistical mechanics.

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22. Annealed Competition of Experts for a Segmentation and Classification of Switching Dynamics

Klaus Pawelzik, Jens Kohlmorgen, Klaus■Robert Müller

https://doi.org/10.1162/neco.1996.8.2.340 | pp. 340-356

We present a method for the unsupervised segmentation of data streams originating from different unknown sources that alternate in time. We use an architecture consisting of competing neural networks. Memory is included to resolve ambiguities of input-output relations. To obtain maximal specialization, the competition is adiabatically increased during training. Our method achieves almost perfect identification and segmentation in the case of switching chaotic dynamics where input manifolds overlap and input-output relations are ambiguous. Only a small dataset is needed for the training procedure. Applications to time series from complex systems demonstrate the potential relevance of our approach for time series analysis and short-term prediction.

23. A Recurrent Network Implementation of Time Series Classification

Vassilios Petridis, Athanasios Kehagias

https://doi.org/10.1162/neco.1996.8.2.357 | pp. 357-372

An incremental credit assignment (ICRA) scheme is introduced and applied to time series classification. It has been inspired from Bayes' rule, but the Bayesian connection is not necessary either for its development or proof of its convergence properties. The ICRA scheme is implemented by a recurrent, hierarchical, modular neural network, which consists of a bank of predictive modules at the lower level, and a decision module at the higher level. For each predictive module, a credit function is computed; the module that best predicts the observed time series behavior receives highest credit. We prove that the credit functions converge (with probability one) to correct values. Simulation results are also presented.

24. Temporal Segmentation in a Neural Dynamic System

D. Horn, Irit Opher

https://doi.org/10.1162/neco.1996.8.2.373 | pp. 373-389

Oscillatory attractor neural networks can perform temporal segmentation, i.e., separate the joint inputs they receive, through the formation of staggered oscillations. This property, which may be basic to many perceptual functions, is investigated here in the context of a symmetric dynamic system. The fully segmented mode is one type of limit cycle that this system can develop. It can be sustained for only a limited number n of oscillators. This limitation to a small number of segments is a basic phenomenon in such systems. Within our model we can explain it in terms of the limited range of narrow subharmonic solutions of the single nonlinear oscillator. Moreover, this point of view allows us to understand the dominance of three leading amplitudes in solutions of partial segmentation, which are obtained for high n. The latter are also abundant when we replace the common input with a graded one, allowing for different inputs to different oscillators. Switching to an input with fluctuating components, we obtain segmentation dominance for small systems and quite irregular waveforms for large systems.

25. Circular Nodes in Neural Networks

Michael Kirby, Rick Miranda

https://doi.org/10.1162/neco.1996.8.2.390 | pp. 390-402

In the usual construction of a neural network, the individual nodes store and transmit real numbers that lie in an interval on the real line; the values are often envisioned as amplitudes. In this article we present a design for a circular node, which is capable of storing and transmitting angular information. We develop the forward and backward propagation formulas for a network containing circular nodes. We show how the use of circular nodes may facilitate the characterization and parameterization of periodic phenomena in general. We describe applications to constructing circular self-maps, periodic compression, and one-dimensional manifold decomposition. We show that a circular node may be used to construct a homeomorphism between a trefoil knot in R3 and a unit circle. We give an application with a network that encodes the dynamic system on the limit cycle of the Kuramoto-Sivashinsky equation. This is achieved by incorporating a circular node in the bottleneck layer of a three-hidden-layer bottleneck network architecture. Exploiting

circular nodes systematically offers a neural network alternative to Fourier series decomposition in approximating periodic or almost periodic functions.

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26. The Computational Power of Discrete Hopfield Nets with Hidden Units

Pekka Orponen

https://doi.org/10.1162/neco.1996.8.2.403 | pp. 403-415

We prove that polynomial size discrete Hopfield networks with hidden units compute exactly the class of Boolean functions PSPACE/poly, i.e., the same functions as are computed by polynomial space-bounded nonuniform Turing machines. As a corollary to the construction, we observe also that networks with polynomially bounded interconnection weights compute exactly the class of functions P/poly, i.e., the class computed by polynomial time-bounded nonuniform Turing machines.

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27. A Self-Organizing Neural Network for the Traveling Salesman Problem That Is Competitive with Simulated Annealing

Marco Budinich

https://doi.org/10.1162/neco.1996.8.2.416 | pp. 416-424

Unsupervised learning applied to an unstructured neural network can give approximate solutions to the traveling salesman problem. For 50 cities in the plane this algorithm performs like the elastic net of Durbin and Willshaw (1987) and it improves when increasing the number of cities to get better than simulated annealing for problems with more than 500 cities. In all the tests this algorithm requires a fraction of the time taken by simulated annealing.

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28. Hierarchical, Unsupervised Learning with Growing via Phase Transitions

David J. Miller, Kenneth Rose

https://doi.org/10.1162/neco.1996.8.2.425 | pp. 425-450

We address unsupervised learning subject to structural constraints, with particular emphasis placed on clustering with an imposed decision tree structure. Most known methods are greedy, optimizing one node of the tree at a time to minimize a local cost. By constrast, we develop a joint optimization method, derived based on information-theoretic principles and closely related to known methods in statistical physics. The approach is inspired by the deterministic annealing algorithm for unstructured data clustering, which was based on maximum entropy inference. The new approach is founded on the principle of minimum cross-entropy, using informative priors to approximate the unstructured clustering solution while imposing the structural constraint. The resulting method incorporates supervised learning principles applied in an unsupervised problem setting. In our approach, the tree “grows” by a sequence of bifurcations that occur while optimizing an effective free energy cost at decreasing temperature scales. Thus, estimates of the tree size and structure are naturally obtained at each temperature in the process. Examples demonstrate considerable improvement over known methods.

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29. The Interchangeability of Learning Rate and Gain in Backpropagation Neural Networks

Georg Thimm, Perry D. Moerland, Emile Fiesler

https://doi.org/10.1162/neco.1996.8.2.451 | pp. 451-460

The backpropagation algorithm is widely used for training multilayer neural networks. In this publication the gain of its activation function(s) is investigated. In specific, it is proven that changing the gain of the activation function is equivalent to changing the learning rate and the weights. This simplifies the backpropagation learning rule by eliminating one of its parameters. The theorem can be extended to hold for some well-known variations on the backpropagation algorithm, such as using a momentum term, flat spot elimination, or adaptive gain. Furthermore, it is successfully applied to compensate for the nonstandard gain of optical sigmoids for optical neural networks.

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30. A Smoothing Regularizer for Feedforward and Recurrent Neural Networks

Lizhong Wu, John Moody

https://doi.org/10.1162/neco.1996.8.3.461 | pp. 461-489

We derive a smoothing regularizer for dynamic network models by requiring robustness in prediction performance to perturbations of the training data. The regularizer can be viewed as a generalization of the first-order Tikhonov stabilizer to dynamic models. For two layer networks with recurrent connections described by the training criterion with the regularizer is where $\Phi = \{U, V, W\}$ is the network parameter set, $Z(t)$ are the targets, $I(t) = \{X(s), s = 1,2, \ldots, t\}$ represents the current and all historical input information, N is the size of the training data set, [Formula: see text] is the regularizer, and λ is a regularization parameter. The closed-form expression for the regularizer for time-lagged recurrent networks is where ■ ■ is the Euclidean matrix norm and γ is a factor that depends upon the maximal value of the first derivatives of the internal unit activations f(). Simplifications of the regularizer are obtained for simultaneous recurrent nets (τ ■ 0), two-layer feedforward nets, and one layer linear nets. We have successfully tested this regularizer in a number of case studies and found that it performs better than standard quadratic weight decay.

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31. Note on the Maxnet Dynamics

John Sum, P.K.S. Tam

https://doi.org/10.1162/neco.1996.8.3.491 | pp. 491-499

A simple method is presented to derive the complete solution of the Maxnet network dynamics. Besides, the exact response time of the network is deduced.

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32. Optimizing Synaptic Conductance Calculation for Network Simulations

William W. Lytton

https://doi.org/10.1162/neco.1996.8.3.501 | pp. 501-509

High computational requirements in realistic neuronal network simulations have led to attempts to realize implementation efficiencies while maintaining as much realism as possible. Since the number of synapses in

a network will generally far exceed the number of neurons, simulation of synaptic activation may be a large proportion of total processing time. We present a consolidating algorithm based on a recent biophysically-inspired simplified Markov model of the synapse. Use of a single lumped state variable to represent a large number of converging synaptic inputs results in substantial speed-ups.

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33. Parameter Extraction from Population Codes: A Critical Assessment

H. P. Snippe

https://doi.org/10.1162/neco.1996.8.3.511 | pp. 511-529

In perceptual systems, a stimulus parameter can be extracted by determining the center-of-gravity of the response profile of a population of neural sensors. Likewise at the motor end of a neural system, center-of-gravity decoding, also known as vector decoding, generates a movement direction from the neural activation profile. We evaluate these schemes from a statistical perspective, by comparing their statistical variance with the minimum variance possible for an unbiased parameter extraction from the noisy neuronal ensemble activation profile. Center-of-gravity decoding can be statistically optimal. This is the case for regular arrays of sensors with gaussian tuning profiles that have an output described by Poisson statistics, and for arrays of sensors with a sinusoidal tuning profile for the (angular) parameter estimated. However, there are also many cases in which center-of-gravity decoding is highly inefficient. This includes the important case where sensor positions are very irregular. Finally, we study the robustness of center-of-gravity decoding against response nonlinearities at different stages of an information processing hierarchy. We conclude that, in neural systems, instead of representing a parameter explicitly, it is safer to leave the parameter coded implicitly in a neuronal ensemble activation profile.

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34. Energy Efficient Neural Codes

William B. Levy, Robert A. Baxter

https://doi.org/10.1162/neco.1996.8.3.531 | pp. 531-543

In 1969 Barlow introduced the phrase "economy of impulses" to express the tendency for successive neural systems to use lower and lower levels of cell firings to produce equivalent encodings. From this viewpoint, the ultimate economy of impulses is a neural code of minimal redundancy. The hypothesis motivating our research is that energy expenditures, e.g., the metabolic cost of recovering from an action potential relative to the cost of inactivity, should also be factored into the economy of impulses. In fact, coding schemes with the largest representational capacity are not, in general, optimal when energy expenditures are taken into account. We show that for both binary and analog neurons, increased energy expenditure per neuron implies a decrease in average firing rate if energy efficient information transmission is to be maintained.

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35. A Nonlinear Hebbian Network that Learns to Detect Disparity in Random-Dot Stereograms

Christopher Lee, Bruno A. Olshausen

https://doi.org/10.1162/neco.1996.8.3.545 | pp. 545-566

An intrinsic limitation of linear, Hebbian networks is that they are capable of learning only from the linear pairwise correlations within an input stream. To explore what higher forms of structure could be learned with a nonlinear Hebbian network, we constructed a model network containing a simple form of nonlinearity and we applied it to the problem of learning to detect the disparities present in random-dot stereograms. The network consists of three layers, with nonlinear sigmoidal activation functions in the second-layer units. The nonlinearities allow the second layer to transform the pixel-based representation in the input layer into a new representation based on coupled pairs of left-right inputs. The third layer of the network then clusters patterns occurring on the second-layer outputs according to their disparity via a standard competitive learning rule. Analysis of the network dynamics shows that the second-layer units' nonlinearities interact with the Hebbian learning rule to expand the region over which pairs of left-right inputs are stable. The learning rule is neurobiologically inspired and plausible, and the model may shed light on how the nervous system learns to use coincidence detection in general.

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36. Coupling the Neural and Physical Dynamics in Rhythmic Movements

Nicholas G. Hatsopoulos

https://doi.org/10.1162/neco.1996.8.3.567 | pp. 567-581

A pair of coupled oscillators simulating a central pattern generator (CPG) interacting with a pendular limb were numerically integrated. The CPG was represented as a van der Pol oscillator and the pendular limb was modeled as a linearized, hybrid spring-pendulum system. The CPG oscillator drove the pendular limb while the pendular limb modulated the frequency of the CPG. Three results were observed. First, sensory feedback influenced the oscillation frequency of the coupled system. The oscillation frequency was lower in the absence of sensory feedback. Moreover, if the muscle gain was decreased, thereby decreasing the oscillation amplitude of the pendular limb and indirectly lowering the effect of sensory feedback, the oscillation frequency decreased monotonically. This is consistent with experimental data (Williamson and Roberts 1986). Second, the CPG output usually led the angular displacement of the pendular limb by a phase of 90 degrees regardless of the length of the limb. Third, the frequency of the coupled system tuned itself to the resonant frequency of the pendular limb. Also, the frequency of the coupled system was highly resistant to changes in the endogenous frequency of the CPG. The results of these simulations support the view that motor behavior emerges from the interaction of the neural dynamics of the nervous system and the physical dynamics of the periphery.

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37. Predictive Minimum Description Length Criterion for Time Series Modeling with Neural Networks

Mikko Lehtokangas, Jukka Saarinen, Pentti Huuhtanen, Kimmo Kaski

https://doi.org/10.1162/neco.1996.8.3.583 | pp. 583-593

Nonlinear time series modeling with a multilayer perceptron network is presented. An important aspect of this modeling is the model selection, i.e., the problem of determining the size as well as the complexity of the model. To overcome this problem we apply the predictive minimum description length (PMDL) principle as a minimization criterion. In the neural network scheme it means minimizing the number of input and hidden units. Three time series modeling experiments are used to examine the usefulness of the PMDL model selection scheme. A comparison with the widely used cross-validation technique is also presented. In our

experiments the PMDL scheme and the cross-validation scheme yield similar results in terms of model complexity. However, the PMDL method was found to be two times faster to compute. This is significant improvement since model selection in general is very time consuming.

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38. Minimum Description Length, Regularization, and Multimodal Data

Richard Rohwer, John C. van der Rest

https://doi.org/10.1162/neco.1996.8.3.595 | pp. 595-609

Relationships between clustering, description length, and regularization are pointed out, motivating the introduction of a cost function with a description length interpretation and the unusual and useful property of having its minimum approximated by the densest mode of a distribution. A simple inverse kinematics example is used to demonstrate that this property can be used to select and learn one branch of a multivalued mapping. This property is also used to develop a method for setting regularization parameters according to the scale on which structure is exhibited in the training data. The regularization technique is demonstrated on two real data sets, a classification problem and a regression problem.

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39. VC Dimension of an Integrate-and-Fire Neuron Model

Anthony M. Zador, Barak A. Pearlmutter

https://doi.org/10.1162/neco.1996.8.3.611 | pp. 611-624

We compute the VC dimension of a leaky integrate-and-fire neuron model. The VC dimension quantifies the ability of a function class to partition an input pattern space, and can be considered a measure of computational capacity. In this case, the function class is the class of integrate-and-fire models generated by varying the integration time constant T and the threshold θ, the input space they partition is the space of continuous-time signals, and the binary partition is specified by whether or not the model reaches threshold at some specified time. We show that the VC dimension diverges only logarithmically with the input signal bandwidth N. We also extend this approach to arbitrary passive dendritic trees. The main contributions of this work are (1) it offers a novel treatment of computational capacity of this class of dynamic system; and (2) it provides a framework for analyzing the computational capabilities of the dynamic systems defined by networks of spiking neurons.

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40. The VC Dimension and Pseudodimension of Two-Layer Neural Networks with Discrete Inputs

Peter L. Bartlett, Robert C. Williamson

https://doi.org/10.1162/neco.1996.8.3.625 | pp. 625-628

We give upper bounds on the Vapnik-Chervonenkis dimension and pseudodimension of two-layer neural networks that use the standard sigmoid function or radial basis function and have inputs from {–D, …,D} n . In Valiant's probably approximately correct (pac) learning framework for pattern classification, and in Haussler's generalization of this framework to nonlinear regression, the results imply that the number of training examples necessary for satisfactory learning performance grows no more rapidly than W log (WD), where W is the number of weights. The previous best bound for these networks was O(W 4).

41. A Theoretical and Experimental Account of n-Tuple Classifier Performance

Richard Rohwer, Micha■ Morciniec

https://doi.org/10.1162/neco.1996.8.3.629 | pp. 629-642

The n-tuple recognition method is briefly reviewed, summarizing the main theoretical results. Large-scale experiments carried out on Stat-Log project datasets confirm this method as a viable competitor to more popular methods due to its speed, simplicity, and accuracy on the majority of a wide variety of classification problems. A further investigation into the failure of the method on certain datasets finds the problem to be largely due to a mismatch between the scales which describe generalization and data sparseness.

42. The Effects of Adding Noise During Backpropagation Training on a Generalization Performance

Guozhong An

https://doi.org/10.1162/neco.1996.8.3.643 | pp. 643-674

We study the effects of adding noise to the inputs, outputs, weight connections, and weight changes of multilayer feedforward neural networks during backpropagation training. We rigorously derive and analyze the objective functions that are minimized by the noise-affected training processes. We show that input noise and weight noise encourage the neural-network output to be a smooth function of the input or its weights, respectively. In the weak-noise limit, noise added to the output of the neural networks only changes the objective function by a constant. Hence, it cannot improve generalization. Input noise introduces penalty terms in the objective function that are related to, but distinct from, those found in the regularization approaches. Simulations have been performed on a regression and a classification problem to further substantiate our analysis. Input noise is found to be effective in improving the generalization performance for both problems. However, weight noise is found to be effective in improving the generalization performance only for the classification problem. Other forms of noise have practically no effect on generalization.

43. Stable Encoding of Large Finite-State Automata in Recurrent Neural Networks with Sigmoid Discriminants

Christian W. Omlin, C. Lee Giles

https://doi.org/10.1162/neco.1996.8.4.675 | pp. 675-696

We propose an algorithm for encoding deterministic finite-state automata (DFAs) in second-order recurrent neural networks with sigmoidal discriminant function and we prove that the languages accepted by the constructed network and the DFA are identical. The desired finite-state network dynamics is achieved by programming a small subset of all weights. A worst case analysis reveals a relationship between the weight strength and the maximum allowed network size, which guarantees finite-state behavior of the constructed network. We illustrate the method by encoding random DFAs with 10, 100, and 1000 states. While the theory predicts that the weight strength scales with the DFA size, we find empirically the weight strength to be almost constant for all the random DFAs. These results can be explained by noting that the generated DFAs represent average cases. We empirically demonstrate the existence of extreme DFAs for which the

weight strength scales with DFA size.

44. Unicycling Helps Your French: Spontaneous Recovery of Associations by Learning Unrelated Tasks

Inman Harvey, James V. Stone

https://doi.org/10.1162/neco.1996.8.4.697 | pp. 697-704

We demonstrate that imperfect recall of a set of associations can usually be improved by training on a new, unrelated set of associations. This spontaneous recovery of associations is a consequence of the high dimensionality of weight spaces, and is therefore not peculiar to any single type of neural net. Accordingly, this work may have implications for spontaneous recovery of memory in the central nervous system.

45. A Theory of the Visual Motion Coding in the Primary Visual Cortex

Li Zhaoping

https://doi.org/10.1162/neco.1996.8.4.705 | pp. 705-730

This paper demonstrates that much of visual motion coding in the primary visual cortex can be understood from a theory of efficient motion coding in a multiscale representation. The theory predicts that cortical cells can have a spectrum of directional indices, be tuned to different directions of motion, and have spatiotemporally separable or inseparable receptive fields (RF). The predictions also include the following correlations between motion coding and spatial, chromatic, and stereo codings: the preferred speed is greater when the cell receptive field size is larger, the color channel prefers lower speed than the luminance channel, and both the optimal speeds and the preferred directions of motion can be different for inputs from different eyes to the same neuron. These predictions agree with experimental observations. In addition, this theory makes predictions that have not been experimentally investigated systematically and provides a testing ground for an efficient multiscale coding framework. These predictions are as follows: (1) if nearby cortical cells of a given preferred orientation and scale prefer opposite directions of motion and have a quadrature RF phase relationship with each other, then they will have the same directional index, (2) a single neuron can have different optimal motion speeds for opposite motion directions of monocular stimuli, and (3) a neuron's ocular dominance may change with motion direction if the neuron prefers opposite directions for inputs from different eyes.

46. Alignment of Coexisting Cortical Maps in a Motor Control Model

Yinong Chen, James A. Reggia

https://doi.org/10.1162/neco.1996.8.4.731 | pp. 731-755

How do multiple feature maps that coexist in the same region of cerebral cortex align with each other? We hypothesize that such alignment is governed by temporal correlations: features in one map that are temporally correlated with those in another come to occupy the same spatial locations in cortex over time. To examine the feasibility of this hypothesis and to establish some of its detailed implications, we studied a multilayered, closed-loop computational model of primary sensorimotor cortex. A simulated arm moving in three dimensions formed the external environment for the model cortical regions. Coexisting proprioceptive

and motor maps formed and generally aligned in a fashion consistent with the temporal correlation hypothesis. For example, in simulated proprioceptive sensory cortex the map of elements responding strongly to stretch of a particular muscle matched the map of tension sensitivity in antagonist muscles. In simulated primary motor cortex the map of elements responding strongly to increased tension in specific muscles matched the map of output elements for the same muscles. These computational results suggest specific experimental measurements that can support or refute the temporal correlation hypothesis for map alignments.

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47. Controlling the Magnification Factor of Self-Organizing Feature Maps

H.-U. Bauer, Ralf Der

https://doi.org/10.1162/neco.1996.8.4.757 | pp. 757-771

The magnification exponents μ occurring in adaptive map formation algorithms like Kohonen's self-organizing feature map deviate for the information theoretically optimal value μ = 1 as well as from the values that optimize, e.g., the mean square distortion error (μ = 1/3 for one-dimensional maps). At the same time, models for categorical perception such as the "perceptual magnet" effect, which are based on topographic maps, require negative magnification exponents μ < 0. We present an extension of the self-organizing feature map algorithm, which utilizes adaptive local learning step sizes to actually control the magnification properties of the map. By change of a single parameter, maps with optimal information transfer, with various minimal reconstruction errors, or with an inverted magnification can be generated. Analytic results on this new algorithm are complemented by numerical simulations.

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48. Semilinear Predictability Minimization Produces Well-Known Feature Detectors

Jürgen Schmidhuber, Martin Eldracher, Bernhard Foltin

https://doi.org/10.1162/neco.1996.8.4.773 | pp. 773-786

Predictability minimization (PM—Schmidhuber 1992) exhibits various intuitive and theoretical advantages over many other methods for unsupervised redundancy reduction. So far, however, there have not been any serious practical applications of PM. In this paper, we apply semilinear PM to static real world images and find that without a teacher and without any significant preprocessing, the system automatically learns to generate distributed representations based on well-known feature detectors, such as orientation-sensitive edge detectors and off-center–on-surround detectors, thus extracting simple features related to those considered useful for image preprocessing and compression.

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49. Learning with Preknowledge: Clustering with Point and Graph Matching Distance Measures

Steven Gold, Anand Rangarajan, Eric Mjolsness

https://doi.org/10.1162/neco.1996.8.4.787 | pp. 787-804

Prior knowledge constraints are imposed upon a learning problem in the form of distance measures. Prototypical 2D point sets and graphs are learned by clustering with point-matching and graph-matching

distance measures. The point-matching distance measure is approximately invariant under affine transformations—translation, rotation, scale, and shear—and permutations. It operates between noisy images with missing and spurious points. The graph-matching distance measure operates on weighted graphs and is invariant under permutations. Learning is formulated as an optimization problem. Large objectives so formulated (~ million variables) are efficiently minimized using a combination of optimization techniques—softassign, algebraic transformations, clocked objectives, and deterministic annealing.

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50. Analog versus Discrete Neural Networks

Bhaskar DasGupta, Georg Schnitger

https://doi.org/10.1162/neco.1996.8.4.805 | pp. 805-818

We show that neural networks with three-times continuously differentiable activation functions are capable of computing a certain family of n-bit boolean functions with two gates, whereas networks composed of binary threshold functions require at least omega(log n) gates. Thus, for a large class of activation functions, analog neural networks can be more powerful than discrete neural networks, even when computing Boolean functions.

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51. On the Relationship between Generalization Error, Hypothesis Complexity, and Sample Complexity for Radial Basis Functions

Partha Niyogi, Federico Girosi

https://doi.org/10.1162/neco.1996.8.4.819 | pp. 819-842

Feedforward networks together with their training algorithms are a class of regression techniques that can be used to learn to perform some task from a set of examples. The question of generalization of network performance from a finite training set to unseen data is clearly of crucial importance. In this article we first show that the generalization error can be decomposed into two terms: the approximation error, due to the insufficient representational capacity of a finite sized network, and the estimation error, due to insufficient information about the target function because of the finite number of samples. We then consider the problem of learning functions belonging to certain Sobolev spaces with gaussian radial basis functions. Using the above-mentioned decomposition we bound the generalization error in terms of the number of basis functions and number of examples. While the bound that we derive is specific for radial basis functions, a number of observations deriving from it apply to any approximation technique. Our result also sheds light on ways to choose an appropriate network architecture for a particular problem and the kinds of problems that can be effectively solved with finite resources, i.e., with a finite number of parameters and finite amounts of data.

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52. Using Neural Networks to Model Conditional Multivariate Densities

Peter M. Williams

https://doi.org/10.1162/neco.1996.8.4.843 | pp. 843-854

Neural network outputs are interpreted as parameters of statistical distributions. This allows us to fit conditional distributions in which the parameters depend on the inputs to the network. We exploit this in modeling multivariate data, including the univariate case, in which there may be input-dependent (e.g.,

time-dependent) correlations between output components. This provides a novel way of modeling conditional correlation that extends existing techniques for determining input-dependent (local) error bars.

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53. Pruning with Replacement on Limited Resource Allocating Networks by F-Projections

Christophe Molina, Mahesan Niranjan

https://doi.org/10.1162/neco.1996.8.4.855 | pp. 855-868

The principle of F-projection, in sequential function estimation, provides a theoretical foundation for a class of gaussian radial basis function networks known as the resource allocating networks (RAN). The ad hoc rules for adaptively changing the size of RAN architectures can be justified from a geometric growth criterion defined in the function space. In this paper, we show that the same arguments can be used to arrive at a pruning with replacement rule for RAN architectures with a limited number of units. We illustrate the algorithm on the laser time series prediction problem of the Santa Fe competition and show that results similar to those of the winners of the competition can be obtained with pruning and replacement.

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54. Engineering Multiversion Neural-Net Systems

Daniel G. Partridge, William B. Yates

https://doi.org/10.1162/neco.1996.8.4.869 | pp. 869-893

In this paper we address the problem of constructing reliable neural-net implementations, given the assumption that any particular implementation will not be totally correct. The approach taken in this paper is to organize the inevitable errors so as to minimize their impact in the context of a multiversion system, i.e., the system functionality is reproduced in multiple versions, which together will constitute the neural-net system. The unique characteristics of neural computing are exploited in order to engineer reliable systems in the form of diverse, multiversion systems that are used together with a "decision strategy" (such as majority vote). Theoretical notions of "methodological diversity" contributing to the improvement of system performance are implemented and tested. An important aspect of the engineering of an optimal system is to overproduce the components and then choose an optimal subset. Three general techniques for choosing final system components are implemented and evaluated. Several different approaches to the effective engineering of complex multiversion systems designs are realized and evaluated to determine overall reliability as well as reliability of the overall system in comparison to the lesser reliability of component substructures.

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55. Biologically Plausible Error-Driven Learning Using Local Activation Differences: The Generalized Recirculation Algorithm

Randall C. O'Reilly

https://doi.org/10.1162/neco.1996.8.5.895 | pp. 895-938

The error backpropagation learning algorithm (BP) is generally considered biologically implausible because it does not use locally available, activation-based variables. A version of BP that can be computed locally using bidirectional activation recirculation (Hinton and McClelland 1988) instead of backpropagated error

derivatives is more biologically plausible. This paper presents a generalized version of the recirculation algorithm (GeneRec), which overcomes several limitations of the earlier algorithm by using a generic recurrent network with sigmoidal units that can learn arbitrary input/output mappings. However, the contrastive Hebbian learning algorithm (CHL, also known as DBM or mean field learning) also uses local variables to perform error-driven learning in a sigmoidal recurrent network. CHL was derived in a stochastic framework (the Boltzmann machine), but has been extended to the deterministic case in various ways, all of which rely on problematic approximations and assumptions, leading some to conclude that it is fundamentally flawed. This paper shows that CHL can be derived instead from within the BP framework via the GeneRec algorithm. CHL is a symmetry-preserving version of GeneRec that uses a simple approximation to the midpoint or second-order accurate Runge-Kutta method of numerical integration, which explains the generally faster learning speed of CHL compared to BI. Thus, all known fully general error-driven learning algorithms that use local activation-based variables in deterministic networks can be considered variations of the GeneRec algorithm (and indirectly, of the backpropagation algorithm). GeneRec therefore provides a promising framework for thinking about how the brain might perform error-driven learning. To further this goal, an explicit biological mechanism is proposed that would be capable of implementing GeneRec-style learning. This mechanism is consistent with available evidence regarding synaptic modification in neurons in the neocortex and hippocampus, and makes further predictions.

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56. Effects of Nonlinear Synapses on the Performance of Multilayer Neural Networks

Günhan Dündar, F-C Hsu, Kenneth Rose

https://doi.org/10.1162/neco.1996.8.5.939 | pp. 939-949

The problems arising from the use of nonlinear multipliers in multilayer neural network synapse structures are discussed. The errors arising from the neglect of nonlinearities are shown and the effect of training in eliminating these errors is discussed. A method for predicting the final errors resulting from nonlinearities is described. Our approximate results are compared with the results from circuit simulations of an actual multiplier circuit.

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57. Modeling Slowly Bursting Neurons via Calcium Store and Voltage-Independent Calcium Current

Teresa Ree Chay

https://doi.org/10.1162/neco.1996.8.5.951 | pp. 951-978

Recent experiments indicate that the calcium store (e.g., endoplasmic reticulum) is involved in electrical bursting and [Ca2+]i oscillation in bursting neuronal cells. In this paper, we formulate a mathematical model for bursting neurons, which includes Ca2+ in the intracellular Ca2+ stores and a voltage-independent calcium channel (VICC). This VICC is activated by a depletion of Ca2+ concentration in the store, [Ca2+]cs. In this model, [Ca2+]cs oscillates slowly, and this slow dynamic in turn gives rise to electrical bursting. The newly formulated model thus is radically different from existing models of bursting excitable cells, whose mechanism owes its origin to the ion channels in the plasma membrane and the [Ca2+]i dynamics. In addition, this model is capable of providing answers to some puzzling phenomena, which the previous models could not (e.g., why cAMP, glucagon, and caffeine have ability to change the burst periodicity).

Using mag-fura-2 fluorescent dyes, it would be interesting to verify the prediction of the model that (1) [Ca2+]cs oscillates in bursting neurons such as Aplysia neuron and (2) the neurotransmitters and hormones that affect the adenylate cyclase pathway can influence this oscillation.

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58. Type I Membranes, Phase Resetting Curves, and Synchrony

Bard Ermentrout

https://doi.org/10.1162/neco.1996.8.5.979 | pp. 979-1001

Type I membrane oscillators such as the Connor model (Connor et al. 1977) and the Morris-Lecar model (Morris and Lecar 1981) admit very low frequency oscillations near the critical applied current. Hansel et al. (1995) have numerically shown that synchrony is difficult to achieve with these models and that the phase resetting curve is strictly positive. We use singular perturbation methods and averaging to show that this is a general property of Type I membrane models. We show in a limited sense that so called Type II resetting occurs with models that obtain rhythmicity via a Hopf bifurcation. We also show the differences between synapses that act rapidly and those that act slowly and derive a canonical form for the phase interactions.

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59. On Neurodynamics with Limiter Function and Linsker's Developmental Model

Jianfeng Feng, Hong Pan, Vwani Roychowdhury

https://doi.org/10.1162/neco.1996.8.5.1003 | pp. 1003-1019

The limiter function is used in many learning and retrieval models as the constraint controlling the magnitude of the weight or state vectors. In this paper, we developed a new method to relate the set of saturated fixed points to the set of system parameters of the models that use the limiter function, and then, as a case study, applied this method to Linsker's Hebbian learning network. We derived a necessary and sufficient condition to test whether a given saturated weight or state vector is stable or not for any given set of system parameters, and used this condition to determine the whole regime in the parameter space over which the given state is stable. This approach allows us to investigate the relative stability of the major receptive fields reported in Linsker's simulations, and to demonstrate the crucial role played by the synaptic density functions.

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60. Effect of Binocular Cortical Misalignment on Ocular Dominance and Orientation Selectivity

Harel Z. Shouval, Nathan Intrator, C. Charles Law, Leon N. Cooper

https://doi.org/10.1162/neco.1996.8.5.1021 | pp. 1021-1040

We model a two-eye visual environment composed of natural images and study its effect on single cell synaptic modification. In particular, we study the effect of binocular cortical misalignment on receptive field formation after eye opening. We show that binocular misalignment affects principal component analysis (PCA) and Bienenstock, Cooper, and Munro (BCM) learning in different ways. For the BCM learning rule this misalignment is sufficient to produce varying degrees of ocular dominance, whereas for PCA learning binocular neurons emerge in every case.

61. A Novel Optimizing Network Architecture with Applications

Anand Rangarajan, Steven Gold, Eric Mjolsness

https://doi.org/10.1162/neco.1996.8.5.1041 | pp. 1041-1060

We present a novel optimizing network architecture with applications in vision, learning, pattern recognition, and combinatorial optimization. This architecture is constructed by combining the following techniques: (1) deterministic annealing, (2) self-amplification, (3) algebraic transformations, (4) clocked objectives, and (5) softassign. Deterministic annealing in conjunction with self-amplification avoids poor local minima and ensures that a vertex of the hypercube is reached. Algebraic transformations and clocked objectives help partition the relaxation into distinct phases. The problems considered have doubly stochastic matrix constraints or minor variations thereof. We introduce a new technique, softassign, which is used to satisfy this constraint. Experimental results on different problems are presented and discussed.

62. Gradient Projection Network: Analog Solver for Linearly Constrained Nonlinear Programming

Kiichi Urahama

https://doi.org/10.1162/neco.1996.8.5.1061 | pp. 1061-1073

An analog approach is presented for solving nonlinear programming problems with linear constraint conditions. The present method is based on transformation of variables with exponential functions, which enables every trajectory to pass through an interior of feasible regions along a gradient direction projected onto the feasible space. Convergence of its trajectory to the solution of optimization problems is guaranteed and it is shown that the present scheme is an extension of the affine scaling method for linear programming to nonlinear programs under a slight modification of Riemannian metric. An analog electronic circuit is also presented that implements the proposed scheme in real time.

63. Online Steepest Descent Yields Weights with Nonnormal Limiting Distribution

Sayandev Mukherjee, Terrence L. Fine

https://doi.org/10.1162/neco.1996.8.5.1075 | pp. 1075-1084

We study the asymptotic properties of the sequence of iterates of weight-vector estimates obtained by training a feedforward neural network with a basic gradient-descent method using a fixed learning rate and no batch-processing. Earlier results based on stochastic approximation techniques (Kuan and Hornik 1991; Finnoff 1993; Bucklew et al. 1993) have established the existence of a gaussian limiting distribution for the weights, but they apply only in the limiting case of a zero learning rate. We here prove, from an exact analysis of the one-dimensional case and constant learning rate, weak convergence to a distribution that is not gaussian in general. We also run simulations to compare and contrast the results of our analysis with those of stochastic approximation.

64. A Numerical Study on Learning Curves in Stochastic Multilayer Feedforward Networks

K. Müller, Michael Finke, Noboru Murata, Klaus Schulten, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.1996.8.5.1085 | pp. 1085-1106

The universal asymptotic scaling laws proposed by Amari et al. are studied in large scale simulations using a CM5. Small stochastic multilayer feedforward networks trained with backpropagation are investigated. In the range of a large number of training patterns t, the asymptotic generalization error scales as 1/t as predicted. For a medium range t a faster 1/t2 scaling is observed. This effect is explained by using higher order corrections of the likelihood expansion. It is shown for small t that the scaling law changes drastically, when the network undergoes a transition from strong overfitting to effective learning.

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65. Rate of Convergence in Density Estimation Using Neural Networks

Dharmendra S. Modha, Elias Masry

https://doi.org/10.1162/neco.1996.8.5.1107 | pp. 1107-1122

Given N i.i.d. observations {X i } N i=1 taking values in a compact subset of R d , such that p * denotes their common probability density function, we estimate p * from an exponential family of densities based on single hidden layer sigmoidal networks using a certain minimum complexity density estimation scheme. Assuming that p * possesses a certain exponential representation, we establish a rate of convergence, independent of the dimension d, for the expected Hellinger distance between the proposed minimum complexity density estimator and the true underlying density p * .

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66. Modeling Conditional Probability Distributions for Periodic Variables

Christopher Bishop, Ian T. Nabney

https://doi.org/10.1162/neco.1996.8.5.1123 | pp. 1123-1133

Most conventional techniques for estimating conditional probability densities are inappropriate for applications involving periodic variables. In this paper we introduce three related techniques for tackling such problems, and investigate their performance using synthetic data. We then apply these techniques to the problem of extracting the distribution of wind vector directions from radar scatterometer data gathered by a remote-sensing satellite.

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67. The Dynamics of Discrete-Time Computation, with Application to Recurrent Neural Networks and Finite State Machine Extraction

M. Casey

https://doi.org/10.1162/neco.1996.8.6.1135 | pp. 1135-1178

Recurrent neural networks (RNNs) can learn to perform finite state computations. It is shown that an RNN performing a finite state computation must organize its state space to mimic the states in the minimal deterministic finite state machine that can perform that computation, and a precise description of the attractor structure of such systems is given. This knowledge effectively predicts activation space dynamics,

which allows one to understand RNN computation dynamics in spite of complexity in activation dynamics. This theory provides a theoretical framework for understanding finite state machine (FSM) extraction techniques and can be used to improve training methods for RNNs performing FSM computations. This provides an example of a successful approach to understanding a general class of complex systems that has not been explicitly designed, e.g., systems that have evolved or learned their internal structure.

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68. Using Bottlenecks in Feedforward Networks as a Dimension Reduction Technique: An Application to Optimization Tasks

Janet Wiles, Paul Bakker, Adam Lynton, Michael Norris, Sean Parkinson, Mark Staples, Alan Whiteside
https://doi.org/10.1162/neco.1996.8.6.1179 | pp. 1179-1183

August 01 1996 Using Bottlenecks in Feedforward Networks as a Dimension Reduction Technique: An Application to Optimization Tasks In Special Collection: CogNet Janet Wiles, Janet Wiles Departments of Computer Science and Psychology, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Paul Bakker, Paul Bakker Departments of Computer Science and Psychology, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Adam Lynton, Adam Lynton Department of Computer Science, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Michael Norris, Michael Norris Department of Computer Science, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Sean Parkinson, Sean Parkinson Department of Computer Science, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Mark Staples, Mark Staples Department of Computer Science, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Alan Whiteside Alan Whiteside Department of Computer Science, The University of Queensland, Queensland 4072, Australia Search for other works by this author on: This Site Google Scholar Author and Article Information Janet Wiles Departments of Computer Science and Psychology, The University of Queensland, Queensland 4072, Australia Paul Bakker Departments of Computer Science and Psychology, The University of Queensland, Queensland 4072, Australia Adam Lynton Department of Computer Science, The University of Queensland, Queensland 4072, Australia Michael Norris Department of Computer Science, The University of Queensland, Queensland 4072, Australia Sean Parkinson Department of Computer Science, The University of Queensland, Queensland 4072, Australia Mark Staples Department of Computer Science, The University of Queensland, Queensland 4072, Australia Alan Whiteside Department of Computer Science, The University of Queensland, Queensland 4072, Australia Received: December 13 1993 Accepted: January 22 1996 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1996 Massachusetts Institute of Technology1996 Neural Computation (1996) 8 (6): 1179–1183. https://doi.org/10.1162/neco.1996.8.6.1179 Article history Received: December 13 1993 Accepted: January 22 1996 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn Email Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Janet Wiles, Paul Bakker, Adam Lynton, Michael Norris, Sean Parkinson, Mark Staples, Alan Whiteside; Using Bottlenecks in Feedforward Networks as a Dimension Reduction Technique: An Application to Optimization Tasks. Neural Comput 1996; 8 (6): 1179–1183. doi: https://doi.org/10.1162/neco.1996.8.6.1179 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural

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69. Temporal Precision of Spike Trains in Extrastriate Cortex of the Behaving Macaque Monkey

Wyeth Bair, Christof Koch

https://doi.org/10.1162/neco.1996.8.6.1185 | pp. 1185-1202

How reliably do action potentials in cortical neurons encode information about a visual stimulus? Most physiological studies do not weigh the occurrences of particular action potentials as significant but treat them only as reflections of average neuronal excitation. We report that single neurons recorded in a previous study by Newsome et al. (1989; see also Britten et al. 1992) from cortical area MT in the behaving monkey respond to dynamic and unpredictable motion stimuli with a markedly reproducible temporal modulation that is precise to a few milliseconds. This temporal modulation is stimulus dependent, being present for highly dynamic random motion but absent when the stimulus translates rigidly.

70. Neural Network for Dynamic Binding with Graph Representation: Form, Linking, and Depth-from-Occlusion

James R. Williamson

https://doi.org/10.1162/neco.1996.8.6.1203 | pp. 1203-1225

A neural network is presented that explicitly represents form attributes and relations between them, thus solving the binding problem without temporal coding. Rather, the network create a graph representation by dynamically allocating nodes to code local form attributes and establishing arcs to link them. With this representation, the network selectively groups and segments in depth objects based on line junction information, producing results consistent with those of several recent visual search experiments. In addition to depth-from-occlusion, the network provides a sufficient framework for local line-labeling processes to recover other three-dimensional (3-D) variables, such as edge/surface contiguity, edge slant, and edge convexity.

71. Neuronal-Based Synaptic Compensation: A Computational Study in Alzheimer's Disease

D. Horn, Nir Levy, Eytan Ruppin

https://doi.org/10.1162/neco.1996.8.6.1227 | pp. 1227-1243

In the framework of an associative memory model, we study the interplay between synaptic deletion and compensation, and memory deterioration, a clinical hallmark of Alzheimer's disease. Our study is motivated by experimental evidence that there are regulatory mechanisms that take part in the homeostasis of neuronal activity and act on the neuronal level. We show that following synaptic deletion, synaptic compensation can be carried out efficiently by a local, dynamic mechanism, where each neuron maintains the profile of its incoming post-synaptic current. Our results open up the possibility that the primary factor in the pathogenesis of cognitive deficiencies in Alzheimer's disease (AD) is the failure of local neuronal

regulatory mechanisms. Allowing for neuronal death, we observe two pathological routes in AD, leading to different correlations between the levels of structural damage and functional decline.

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72. Spike Train Processing by a Silicon Neuromorph: The Role of Sublinear Summation in Dendrites

D.P.M. Northmore, John G. Elias

https://doi.org/10.1162/neco.1996.8.6.1245 | pp. 1245-1265

A dendritic tree, as part of a silicon neuromorph, was modeled in VLSI as a multibranched, passive cable structure with multiple synaptic sites that either depolarize or hyperpolarize local "membrane patches," thereby raising or lowering the probability of spike generation of an integrate-and-fire "soma." As expected from previous theoretical analyses, contemporaneous synaptic activation at widely separated sites on the artificial tree resulted in near-linear summation, as did neighboring excitatory and inhibitory activations. Activation of synapses of the same type close in time and space produced local saturation of potential, resulting in spike train processing capabilities not possible with linear summation alone. The resulting sublinear synaptic summation, as well as being physiologically plausible, is sufficient for a variety of spike train processing functions. With the appropriate arrangement of synaptic inputs on its dendritic tree, a neuromorph was shown to discriminate input pulse intervals and patterns, pulse train frequencies, and detect correlation between input trains.

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73. A Large Committee Machine Learning Noisy Rules

Robert Urbanczik

https://doi.org/10.1162/neco.1996.8.6.1267 | pp. 1267-1276

Statistical mechanics is used to study generalization in a tree committee machine with K hidden units and continuous weights trained on examples generated by a teacher of the same structure but corrupted by noise. The corruption is due to additive gaussian noise applied in the input layer or the hidden layer of the teacher. In the large K limit the generalization error ε g as function of α, the number of patterns per adjustable parameter, shows a qualitatively similar behavior for the two cases: It does not approach its optimal value and is nonmonotonic if training is done at zero temperature. This remains true even when replica symmetry breaking is taken into account. Training at a fixed positive temperature leads, within the replica symmetric theory, to an α -k decay of ε g toward its optimal value. The value of k is calculated and found to depend on the model of noise. By scaling the temperature with α, the value of k can be increased to an optimal value k opt . However, at one step of replica symmetry breaking at a fixed positive temperature ε g decays as α –k opt . So, although ε g will approach its optimal value with increasing sample size for any fixed K, the convergence is only uniform in K when training at a positive temperature.

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74. Vapnik-Chervonenkis Generalization Bounds for Real Valued Neural Networks

Arne Hole

https://doi.org/10.1162/neco.1996.8.6.1277 | pp. 1277-1299

We show how lower bounds on the generalization ability of feedforward neural nets with real outputs can be derived within a formalism based directly on the concept of VC dimension and Vapnik's theorem on uniform convergence of estimated probabilities.

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75. The Error Surface of the Simplest XOR Network Has Only Global Minima

I.G. Sprinkhuizen-Kuyper, Egbert J. W. Boers

https://doi.org/10.1162/neco.1996.8.6.1301 | pp. 1301-1320

The artificial neural network with one hidden unit and the input units connected to the output unit is considered. It is proven that the error surface of this network for the patterns of the XOR problem has minimum values with zero error and that all other stationary points of the error surface are saddlepoints. Also, the volume of the regions in weight space with saddlepoints is zero, hence training this network on the four patterns of the XOR problem using, e.g., backpropagation with momentum, the correct solution with error zero will be reached in the limit with probability one.

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76. Statistical Approach to Shape from Shading: Reconstruction of Three-Dimensional Face Surfaces from Single Two-Dimensional Images

Joseph J. Atick, Paul A. Griffin, Amanda Redlich

https://doi.org/10.1162/neco.1996.8.6.1321 | pp. 1321-1340

The human visual system is proficient in perceiving three-dimensional shape from the shading patterns in a two-dimensional image. How it does this is not well understood and continues to be a question of fundamental and practical interest. In this paper we present a new quantitative approach to shape-from-shading that may provide some answers. We suggest that the brain, through evolution or prior experience, has discovered that objects can be classified into lower-dimensional object-classes as to their shape. Extraction of shape from shading is then equivalent to the much simpler problem of parameter estimation in a low-dimensional space. We carry out this proposal for an important class of three-dimensional (3D) objects: human heads. From an ensemble of several hundred laser-scanned 3D heads, we use principal component analysis to derive a low-dimensional parameterization of head shape space. An algorithm for solving shape-from-shading using this representation is presented. It works well even on real images where it is able to recover the 3D surface for a given person, maintaining facial detail and identity, from a single 2D image of his face. This algorithm has applications in face recognition and animation.

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77. The Lack of A Priori Distinctions Between Learning Algorithms

David H. Wolpert

https://doi.org/10.1162/neco.1996.8.7.1341 | pp. 1341-1390

This is the first of two papers that use off-training set (OTS) error to investigate the assumption-free relationship between learning algorithms. This first paper discusses the senses in which there are no a priori distinctions between learning algorithms. (The second paper discusses the senses in which there are such distinctions.) In this first paper it is shown, loosely speaking, that for any two algorithms A and B, there are

“as many” targets (or priors over targets) for which A has lower expected OTS error than B as vice versa, for loss functions like zero-one loss. In particular, this is true if A is cross-validation and B is “anti-cross-validation” (choose the learning algorithm with largest cross-validation error). This paper ends with a discussion of the implications of these results for computational learning theory. It is shown that one cannot say: if empirical misclassification rate is low, the Vapnik-Chervonenkis dimension of your generalizer is small, and the training set is large, then with high probability your OTS error is small. Other implications for “membership queries” algorithms and “punting” algorithms are also discussed.

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78. The Existence of A Priori Distinctions Between Learning Algorithms

David H. Wolpert

https://doi.org/10.1162/neco.1996.8.7.1391 | pp. 1391-1420

This is the second of two papers that use off-training set (OTS) error to investigate the assumption-free relationship between learning algorithms. The first paper discusses a particular set of ways to compare learning algorithms, according to which there are no distinctions between learning algorithms. This second paper concentrates on different ways of comparing learning algorithms from those used in the first paper. In particular this second paper discusses the associated a priori distinctions that do exist between learning algorithms. In this second paper it is shown, loosely speaking, that for loss functions other than zero-one (e.g., quadratic loss), there are a priori distinctions between algorithms. However, even for such loss functions, it is shown here that any algorithm is equivalent on average to its “randomized” version, and in this still has no first principles justification in terms of average error. Nonetheless, as this paper discusses, it may be that (for example) cross-validation has better head-to-head minimax properties than “anti-cross-validation” (choose the learning algorithm with the largest cross-validation error). This may be true even for zero-one loss, a loss function for which the notion of “randomization” would not be relevant. This paper also analyzes averages over hypotheses rather than targets. Such analyses hold for all possible priors over targets. Accordingly they prove, as a particular example, that cross-validation cannot be justified as a Bayesian procedure. In fact, for a very natural restriction of the class of learning algorithms, one should use anti-cross-validation rather than cross-validation (!).

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79. No Free Lunch for Cross-Validation

Huaiyu Zhu, Richard Rohwer

https://doi.org/10.1162/neco.1996.8.7.1421 | pp. 1421-1426

It is known theoretically that an algorithm cannot be good for an arbitrary prior. We show that in practical terms this also applies to the technique of “cross-validation,” which has been widely regarded as defying this general rule. Numerical examples are analyzed in detail. Their implications to researches on learning algorithms are discussed.

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80. A Self-Organizing Model of “Color Blob” Formation

Harry G. Barrow, Alistair J. Bray, Julian M. L. Budd

https://doi.org/10.1162/neco.1996.8.7.1427 | pp. 1427-1448

This paper explores the possibility that the formation of color blobs in primate striate cortex can be partly explained through the process of activity-based self-organization. We present a simulation of a highly simplified model of visual processing along the parvocellular pathway, that combines precortical color processing, excitatory and inhibitory cortical interactions, and Hebbian learning. The model self-organizes in response to natural color images and develops islands of unoriented, color-selective cells within a sea of contrast-sensitive, orientation-selective cells. By way of understanding this topography, a principal component analysis of the color inputs presented to the network reveals that the optimal linear coding of these inputs keeps color information and contrast information separate.

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81. Functional Consequences of an Integration of Motion and Stereopsis in Area MT of Monkey Extrastriate Visual Cortex

Markus Lappe

https://doi.org/10.1162/neco.1996.8.7.1449 | pp. 1449-1461

Experimental evidence from neurophysiological recordings in the middle temporal (MT) area of the macaque monkey suggests that motion-selective cells can use disparity information to separate motion signals that originate from different depths. This finding of a cross-talk between different visual channels has implications for the understanding of the processing of motion in the primate visual system and especially for behavioral tasks requiring the determination of global motion. In this paper, the consequences for the analysis of optic flow fields are explored. A network model is presented that effectively uses the disparity sensitivity of MT-like neurons for the reduction of noise in optic flow fields. Simulations reproduce the recent psychophysical finding that the robustness of the human optic flow processing system is improved by stereoscopic depth information, but that the use of this information depends on the structure of the visual environment.

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82. Learning Perceptually Salient Visual Parameters Using Spatiotemporal Smoothness Constraints

James V. Stone

https://doi.org/10.1162/neco.1996.8.7.1463 | pp. 1463-1492

A model is presented for unsupervised learning of low level vision tasks, such as the extraction of surface depth. A key assumption is that perceptually salient visual parameters (e.g., surface depth) vary smoothly over time. This assumption is used to derive a learning rule that maximizes the long-term variance of each unit's outputs, whilst simultaneously minimizing its short-term variance. The length of the half-life associated with each of these variances is not critical to the success of the algorithm. The learning rule involves a linear combination of anti-Hebbian and Hebbian weight changes, over short and long time scales, respectively. This maximizes the information throughput with respect to low-frequency parameters implicit in the input sequence. The model is used to learn stereo disparity from temporal sequences of random-dot and gray-level stereograms containing synthetically generated subpixel disparities. The presence of temporal discontinuities in disparity does not prevent learning or generalization to previously unseen image sequences. The implications of this class of unsupervised methods for learning in perceptual systems are discussed.

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83. Using Visual Latencies to Improve Image Segmentation

Ralf Opara, Florentin Wörgötter

https://doi.org/10.1162/neco.1996.8.7.1493 | pp. 1493-1520

An artificial neural network model is proposed that combines several aspects taken from physiological observations (oscillations, synchronizations) with a visual latency mechanism in order to achieve an improved analysis of visual scenes. The network consists of two parts. In the lower layers that contain no lateral connections the propagation velocity of the activity of the units depends on the contrast of the individual objects in the scene. In the upper layers lateral connections are used to achieve synchronization between corresponding image parts. This architecture assures that the activity that arises in response to a scene containing objects with different contrast is spread out over several layers in the network. Thereby adjacent objects with different contrast will be separated and synchronization occurs in the upper layers without mutual disturbance between different objects. A comparison with a one-layer network shows that synchronization occurs in the upper layers without mutual disturbance between different objects. A comparison with a one-layer network shows that synchronization in the latency dependent multilayer net is indeed achieved much faster as soon as more than five objects have to be recognized. In addition, it is shown that the network is highly robust against noise in the stimuli or variations in the propagation delays (latencies), respectively. For a consistent analysis of a visual scene the different features of an individual object have to be recognized as belonging together and separated from other objects. This study shows that temporal differences, naturally introduced by stimulus latencies in every biological sensory system, can strongly improve the performance and allow for an analysis of more complex scenes.

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84. Learning and Generalization in Cascade Network Architectures

Enno Littmann, Helge Ritter

https://doi.org/10.1162/neco.1996.8.7.1521 | pp. 1521-1539

Incrementally constructed cascade architectures are a promising alternative to networks of predefined size. This paper compares the direct cascade architecture (DCA) proposed in Littmann and Ritter (1992) to the cascade-correlation approach of Fahlman and Lebiere (1990) and to related approaches and discusses the properties on the basis of various benchmark results. One important virtue of DCA is that it allows the cascading of entire subnetworks, even if these admit no error-backpropagation. Exploiting this flexibility and using LLM networks as cascaded elements, we show that the performance of the resulting network cascades can be greatly enhanced compared to the performance of a single network. Our results for the Mackey-Glass time series prediction task indicate that such deeply cascaded network architectures achieve good generalization even on small data sets, when shallow, broad architectures of comparable size suffer from overfitting. We conclude that the DCA approach offers a powerful and flexible alternative to existing schemes such as, e.g., the mixtures of experts approach, for the construction of modular systems from a wide range of subnetwork types.

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85. Hybrid Modeling, HMM/NN Architectures, and Protein Applications

Pierre Baldi, Yves Chauvin

https://doi.org/10.1162/neco.1996.8.7.1541 | pp. 1541-1565

We describe a hybrid modeling approach where the parameters of a mode are calculated and modulated by another model, typically a neural network (NN), to avoid both overfitting and underfitting. We develop the approach for the case of Hidden Markov Models (HMMs), by deriving a class of hybrid HMM/NN architectures. These architectures can be trained with unified algorithms that blend HMM dynamic programming with NN backpropagation. In the case of complex data, mixtures of HMMs or modulated HMMs must be used. NNs can then be applied both to the parameters of each single HMM, and to the switching or modulatation of the models, as a function of input or context. Hybrid HMM/NN architectures provide a flexible NN parameterization for the control of model structure and complexity. At the same time, they can capture distributions that, in practice, are inaccessible to single HMMs. The HMM/NN hybrid approach is tested, in its simplest form, by constructing a model of the immunoglobulin protein family. A hybrid model is trained, and a multiple alignment derived, with less than a fourth of the number of parameters used with previous single HMMs.

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86. Synchronized Action of Synaptically Coupled Chaotic Model Neurons

Henry D. I. Abarbanel, Ramón Huerta, M. I. Rabinovich, Nikolai F. Rulkov, Peter F. Rowat, Allen I. Selverston

https://doi.org/10.1162/neco.1996.8.8.1567 | pp. 1567-1602

Experimental observations of the intracellular recorded electrical activity in individual neurons show that the temporal behavior is often chaotic. We discuss both our own observations on a cell from the stomatogastric central pattern generator of lobster and earlier observations in other cells. In this paper we work with models with chaotic neurons, building on models by Hindmarsh and Rose for bursting, spiking activity in neurons. The key feature of these simplified models of neurons is the presence of coupled slow and fast subsystems. We analyze the model neurons using the same tools employed in the analysis of our experimental data. We couple two model neurons both electrotonically and electrochemically in inhibitory and excitatory fashions. In each of these cases, we demonstrate that the model neurons can synchronize in phase and out of phase depending on the strength of the coupling. For normal synaptic coupling, we have a time delay between the action of one neuron and the response of the other. We also analyze how the synchronization depends on this delay. A rich spectrum of synchronized behaviors is possible for electrically coupled neurons and for inhibitory coupling between neurons. In synchronous neurons one typically sees chaotic motion of the coupled neurons. Excitatory coupling produces essentially periodic voltage trajectories, which are also synchronized. We display and discuss these synchronized behaviors using two "distance" measures of the synchronization.

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87. On the Capacity of Threshold Adalines with Limited-Precision Weights

M. Stevenson, Shaheedul Huq

https://doi.org/10.1162/neco.1996.8.8.1603 | pp. 1603-1610

The effect of limited-precision weights on the capacity of the threshold Adaline is examined. In particular, an experimental technique is used to determine the capacity of the threshold Adaline for several different levels of weight precision. The results provide some insight into the manner in which the capacity of the threshold Adaline increases as the number of bits per weight increases. As might be expected, the growth in capacity due to an additional bit per weight is greatest when the original number of bits per weight is small, and

decreases as the original number of bits per weight increases.

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88. Binocular Receptive Field Models, Disparity Tuning, and Characteristic Disparity

Yudong Zhu, Ning Qian

https://doi.org/10.1162/neco.1996.8.8.1611 | pp. 1611-1641

Disparity tuning of visual cells in the brain depends on the structure of their binocular receptive fields (RFs). Freeman and coworkers have found that binocular RFs of a typical simple cell can be quantitatively described by two Gabor functions with the same gaussian envelope but different phase parameters in the sinusoidal modulations (Freeman and Ohzawa 1990). This phase-parameter-based RF description has recently been questioned by Wagner and Frost (1993) based on their identification of a so-called characteristic disparity (CD) in some cells' disparity tuning curves. They concluded that their data favor the traditional binocular RF model, which assumes on overall positional shift between a cell's left and right RFs. Here we set to resolve this issue by studying the dependence of cells' disparity tuning on their underlying RF structures through mathematical analyses and computer simulations. We model the disparity tuning curves in Wagner and Frost's experiments and demonstrate that the mere existence of approximate CDs in real cells cannot be used to distinguish the phase-parameter-based RF description from the traditional position-shift-based RF description. Specifically, we found that model simple cells with either type RF description do not have a CD. Model complex cells with the position-shift-based RF description have a precise CD, and those with the phase-parameter-based RF description have an approximate CD. We also suggest methods for correctly distinguishing the two types of RF descriptions. A hybrid of the two RF models may be required to fit the behavior of some real cells, and we show how to determine the relative contributions of the two RF models.

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89. Response Characteristics of a Low-Dimensional Model Neuron

Bo Cartling

https://doi.org/10.1162/neco.1996.8.8.1643 | pp. 1643-1652

It is shown that a low-dimensional model neuron with a response time constant smaller than the membrane time constant closely reproduces the activity and excitability behavior of a detailed conductance-based model of Hodgkin-Huxley type. The fast response of the activity variable also makes it possible to reduce the model to a one-dimensional model, in particular for typical conditions. As an example, the reduction to a single-variable model from a multivariable conductance-based model of a neocortical pyramidal cell with somatic input is demonstrated. The conditions for avoiding a spurious damped oscillatory response to a constant input are derived, and it is shown that a limit-cycle response cannot occur. The capability of the low-dimensional model to approximate higher-dimensional models accurately makes it useful for describing complex dynamics of nets of interconnected neurons. The simplicity of the model facilitates analytic studies, elucidations of neurocomputational mechanisms, and applications to large-scale systems.

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90. What Matters in Neuronal Locking?

Wulfram Gerstner, J. Leo van Hemmen, Jack D. Cowan
https://doi.org/10.1162/neco.1996.8.8.1653 | pp. 1653-1676

Exploiting local stability, we show what neuronal characteristics are essential to ensure that coherent oscillations are asymptotically stable in a spatially homogeneous network of spiking neurons. Under standard conditions, a necessary and, in the limit of a large number of interacting neighbors, also sufficient condition is that the postsynaptic potential is increasing in time as the neurons fire. If the postsynaptic potential is decreasing, oscillations are bound to be unstable. This is a kind of locking theorem and boils down to a subtle interplay of axonal delays, postsynaptic potentials, and refractory behavior. The theorem also allows for mixtures of excitatory and inhibitory interactions. On the basis of the locking theorem, we present a simple geometric method to verify the existence and local stability of a coherent oscillation.

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91. Hebbian Learning of Context in Recurrent Neural Networks

Nicolas Brunel
https://doi.org/10.1162/neco.1996.8.8.1677 | pp. 1677-1710

Single electrode recording in the inferotemporal cortex of monkeys during delayed visual memory tasks provide evidence for attractor dynamics in the observed region. The persistent elevated delay activities could be internal representations of features of the learned visual stimuli shown to the monkey during training. When uncorrelated stimuli are presented during training in a fixed sequence, these experiments display significant correlations between the internal representations. Recently a simple model of attractor neural network has reproduced quantitatively the measured correlations. An underlying assumption of the model is that the synaptic matrix formed during the training phase contains in its efficacies information about the contiguity of persistent stimuli in the training sequence. We present here a simple unsupervised learning dynamics that produces such a synaptic matrix if sequences of stimuli are repeatedly presented to the network at fixed order. The resulting matrix is then shown to convert temporal correlations during training into spatial correlations between attractors. The scenario is that, in the presence of selective delay activity, at the presentation of each stimulus, the activity distribution in the neural assembly contain information of both the current stimulus and the previous one (carried by the attractor). Thus the recurrent synaptic matrix can code not only for each of the stimuli presented to the network but also for their context. We combine the idea that for learning to be effective, synaptic modification should be stochastic, with the fact that attractors provide learnable information about two consecutive stimuli. We calculate explicitly the probability distribution of synaptic efficacies as a function of training protocol, that is, the order in which stimuli are presented to the network. We then solve for the dynamics of a network composed of integrate-and-fire excitatory and inhibitory neurons with a matrix of synaptic collaterals resulting from the learning dynamics. The network has a stable spontaneous activity, and stable delay activity develops after a critical learning stage. The availability of a learning dynamics makes possible a number of experimental predictions for the dependence of the delay activity distributions and the correlations between them, on the learning stage and the learning protocol. In particular it makes specific predictions for pair-associates delay experiments.

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92. Neural Correlation via Random Connections

Joshua Chover
https://doi.org/10.1162/neco.1996.8.8.1711 | pp. 1711-1729

A simple neural network is studied, which has sparse, random, plastic, excitatory connections and also feedback loops between sensory cells and correlator cells. Time is limited to several discrete instants, where firing is synchronous. For parameter values within biological ranges, the system exhibits a capacity for associative recall, with a controlled amount of extraneous firing, following Hebb-like synaptic changes.

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93. Singular Perturbation Analysis of Competitive Neural Networks with Different Time Scales

Anke Meyer■Baese, Frank W. Ohl, Henning Scheich

https://doi.org/10.1162/neco.1996.8.8.1731 | pp. 1731-1742

The dynamics of complex neural networks must include the aspects of long- and short-term memory. The behavior of the network is characterized by an equation of neural activity as a fast phenomenon and an equation of synaptic modification as a slow part of the neural system. The main idea of this paper is to apply a stability analysis method of fixed points of the combined activity and weight dynamics for a special class of competitive neural networks. We present a quadratic-type Lyapunov function for the flow of a competitive neural system with fast and slow dynamic variables as a global stability method and a modality of detecting the local stability behavior around individual equilibrium points.

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94. How Dependencies between Successive Examples Affect On-Line Learning

Wim Wiegerinck, Tom Heskes

https://doi.org/10.1162/neco.1996.8.8.1743 | pp. 1743-1765

We study the dynamics of on-line learning for a large class of neural networks and learning rules, including backpropagation for multilayer perceptrons. In this paper, we focus on the case where successive examples are dependent, and we analyze how these dependencies affect the learning process. We define the representation error and the prediction error. The representation error measures how well the environment is represented by the network after learning. The prediction error is the average error that a continually learning network makes on the next example. In the neighborhood of a local minimum of the error surface, we calculate these errors. We find that the more predictable the example presentation, the higher the representation error, i.e., the less accurate the asymptotic representation of the whole environment. Furthermore we study the learning process in the presence of a plateau. Plateaus are flat spots on the error surface, which can severely slow down the learning process. In particular, they are notorious in applications with multilayer perceptrons. Our results, which are confirmed by simulations of a multilayer perceptron learning a chaotic time series using backpropagation, explain how dependencies between examples can help the learning process to escape from a plateau.

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95. Autonomous Design of Artificial Neural Networks by Neurex

François Michaud, Ruben Gonzalez Rubio

https://doi.org/10.1162/neco.1996.8.8.1767 | pp. 1767-1786

Artificial neural networks (ANN) have been demonstrated to be increasingly more useful for complex problems difficult to solve with conventional methods. With their learning abilities, they avoid having to develop a mathematical model or acquiring the appropriate knowledge to solve a task. The difficulty now lies in the ANN design process. A lot of choices must be made to design an ANN, and there are no available design rules to make these choices directly for a particular problem. Therefore, the design for an ANN demands a certain number of iterations, mainly guided by the expertise and the intuition of the developer. To automate the ANN design process, we have developed Neurex, composed of an expert system and an ANN simulator. Neurex autonomously guides the iterative ANN design process. Its structure tries to reproduce the design steps done by a human expert in conceiving an ANN. As a whole, the Neurex structure serves as a framework to implement this expertise for different learning paradigms. This article presents the system's general characteristics and its use in designing ANN using the standard backpropagation learning law.

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96. Index

(authors unknown)

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